

MGnify

The microbiome sequence data analysis resource
An introduction to metagenomics and amplicon
analyses

Tanya Gurbich, Germana Baldi

Bioinformaticians

tgurbich@ebi.ac.uk
germanab@ebi.ac.uk



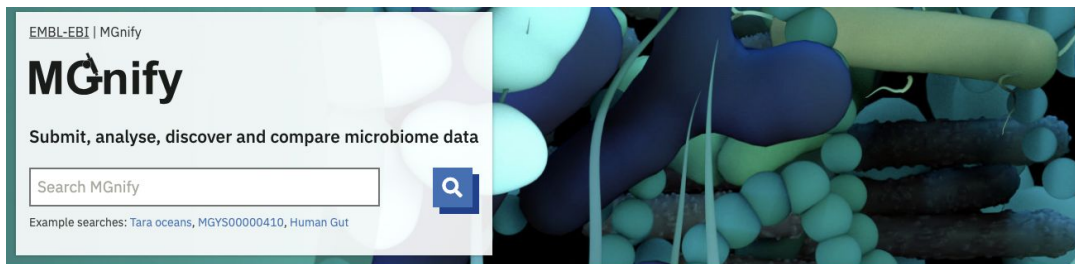
Schedule for the day

- Why should we study microbiomes?
- Different approaches to studying microbiomes
- Culture-independent approaches:
 - Amplicon (lecture)
 - Amplicon (practical)
 - Metagenomics (lecture)
 - Metagenomics (practical)
- Lunch
- The MGnify resource
- Future directions

Who we are



Who we are

The header of the MGnify website. On the left, there is a white box with the EMBL-EBI logo and the text 'MGnify'. Below this, it says 'Submit, analyse, discover and compare microbiome data'. There is a search bar with the placeholder text 'Search MGnify' and a magnifying glass icon. Below the search bar, it says 'Example searches: Tara oceans, MGYS00000410, Human Gut'. To the right of the white box is a large, colorful, abstract image of a microbiome, showing various shapes and colors like blue, green, and yellow.

A resource for assembly, analysis and archiving of microbiome-derived sequencing data

Overview Submit data ▼ Text search ▼ Browse data ▼ MGnify Proteins ▼ MGnify Genomes ▼ Help ▼ Login

Search by

Text search →

Name, biome, or keyword

Sequence search →

Sequence search

Or by data type

xxx Analysis types

507750 amplicon

62596 assemblies

2050 metabarcoding

40250 metagenomes

2581 metatranscriptomics

2 long reads
assemblies

Public data

5129 studies

629821 analyses

518782 genomes in 19
MAG catalogues

Request analysis of

Submit and/or Request →

Your data

Request →

A public dataset

Latest studies



Analise de metais pesado de rios

Analise de biodiversidade de areas contaminadas por metais pesados



MicroReset rabbit

Characterization of the rabbit (*Oryctolagus cuniculus*) fecal metagenome by deep shotgun sequencing. The MicroReset Project was funded by the Animal Health program of the Carnot Institute in France. Sequencing data was generated by Genotoul (Toulouse,...



EMG produced TPA metagenomics assembly of PRJNA944553 data set (Metagenomic samples of cat and human...

The Third Party Annotation (TPA) assembly was derived from the primary whole

Who we are



What is a microbiome and how is it relevant to plants?



What is a microbiome?

The microbiome is the **community of microorganisms** (such as fungi, bacteria and viruses) that exists in a **particular environment**.

The role of microbiome in crop health and plant breeding

- Plants interact with a diverse microbial community
- Microbiomes influence plant health, growth, and productivity

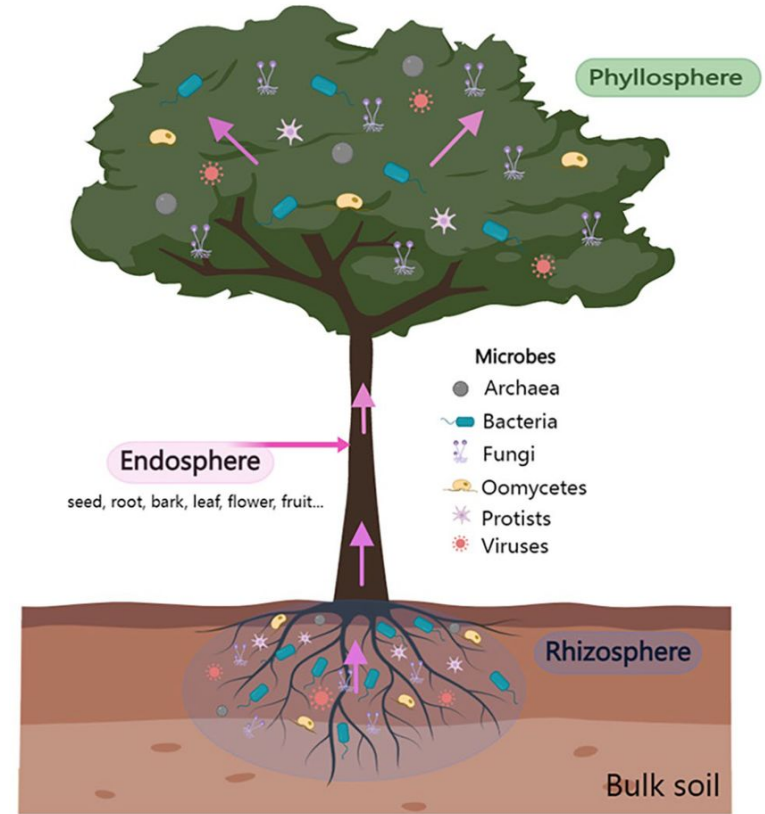


Image from DOI:10.3390/ijms242216118

The role of microbiome in crop health and plant breeding

- Nutrient acquisition
 - Nitrogen fixation: *Rhizobium* in legumes → symbiotic N supply
 - Phosphate solubilisation: *Pseudomonas* and *Bacillus* make soil phosphate bioavailable

The role of microbiome in crop health and plant breeding

- Nutrient acquisition
 - Nitrogen fixation: *Rhizobium* in legumes → symbiotic N supply
 - Phosphate solubilisation: *Pseudomonas* and *Bacillus* make soil phosphate bioavailable
- Disease resistance
 - Antibiotic-producing bacteria: *Streptomyces* in the rhizosphere suppress fungal pathogens
 - Induced systemic resistance (ISR): *Pseudomonas fluorescens* primes plant immunity

The role of microbiome in crop health and plant breeding

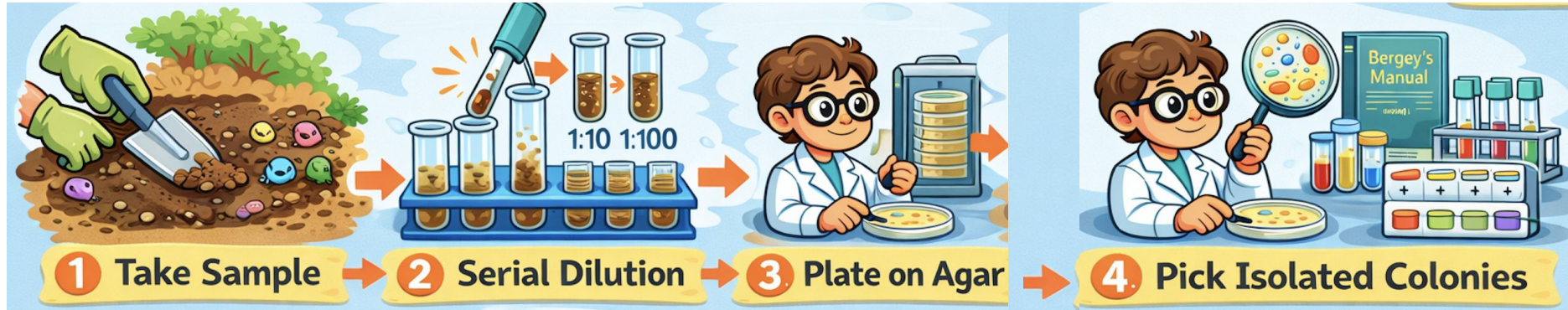
- Nutrient acquisition
 - Nitrogen fixation: *Rhizobium* in legumes → symbiotic N supply
 - Phosphate solubilisation: *Pseudomonas* and *Bacillus* make soil phosphate bioavailable
- Disease resistance
 - Antibiotic-producing bacteria: *Streptomyces* in the rhizosphere suppress fungal pathogens
 - Induced systemic resistance (ISR): *Pseudomonas fluorescens* primes plant immunity
- Abiotic stress tolerance
 - Drought tolerance: *Bacillus subtilis* and *Arthrobacter* improve water-use efficiency
 - Salt tolerance: Endophytic *Enterobacter* species reduce salt stress in rice

The role of microbiome in crop health and plant breeding

- Nutrient acquisition
 - Nitrogen fixation: *Rhizobium* in legumes → symbiotic N supply
 - Phosphate solubilisation: *Pseudomonas* and *Bacillus* make soil phosphate bioavailable
- Disease resistance
 - Antibiotic-producing bacteria: *Streptomyces* in the rhizosphere suppress fungal pathogens
 - Induced systemic resistance (ISR): *Pseudomonas fluorescens* primes plant immunity
- Abiotic stress tolerance
 - Drought tolerance: *Bacillus subtilis* and *Arthrobacter* improve water-use efficiency
 - Salt tolerance: Endophytic *Enterobacter* species reduce salt stress in rice
- Plant growth promotion
 - Hormone production: *Azospirillum* produces auxins, promoting root development

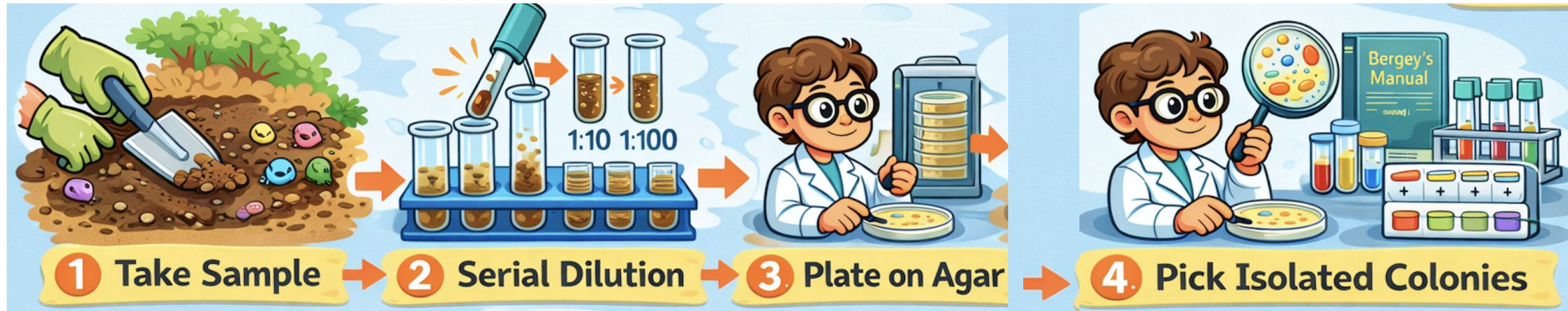
Approaches to study the microbiome

The traditional route: isolate the microorganisms



Approaches to study the microbiome

The traditional route: isolate the microorganisms



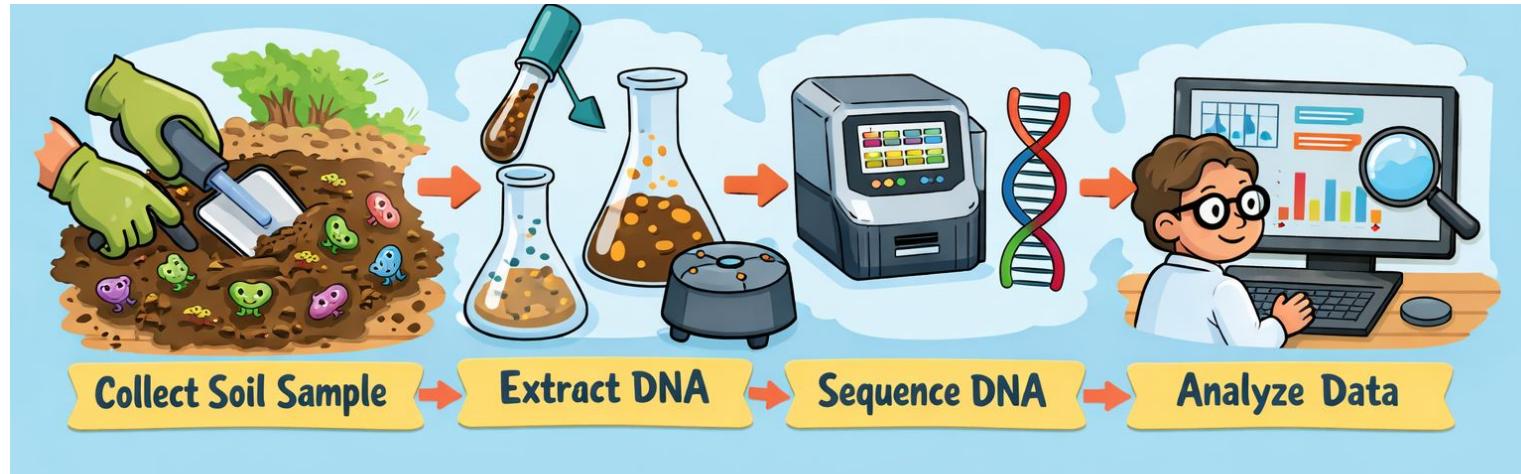
- Between 10 billion and 1 trillion microbial cells in just 1 gram of soil
- Thousands species per 1 gram of soil
- Not all bacteria will grow in the lab

What is culture-independent sequencing?

Functional and sequence-based analysis of the collective microbial genomes of an environmental sample

The goal is to completely characterise a microbial community:

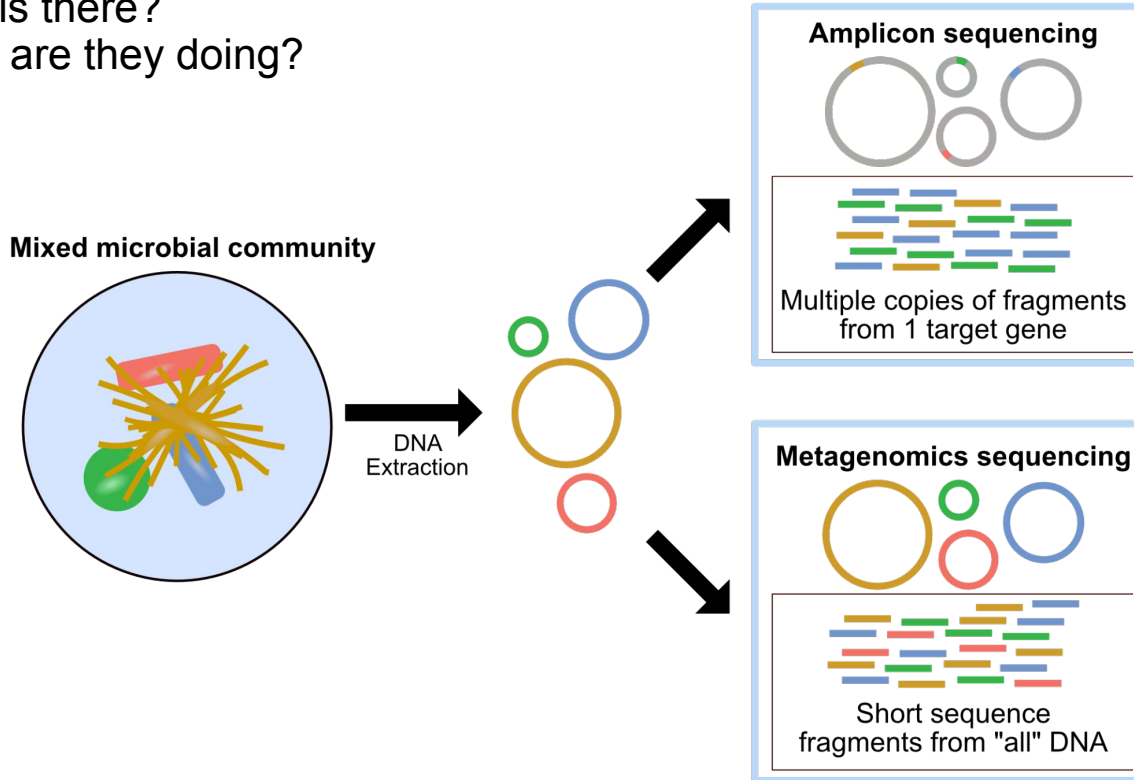
- Who is there?
- What are they doing?



Amplicon vs metagenomic sequencing

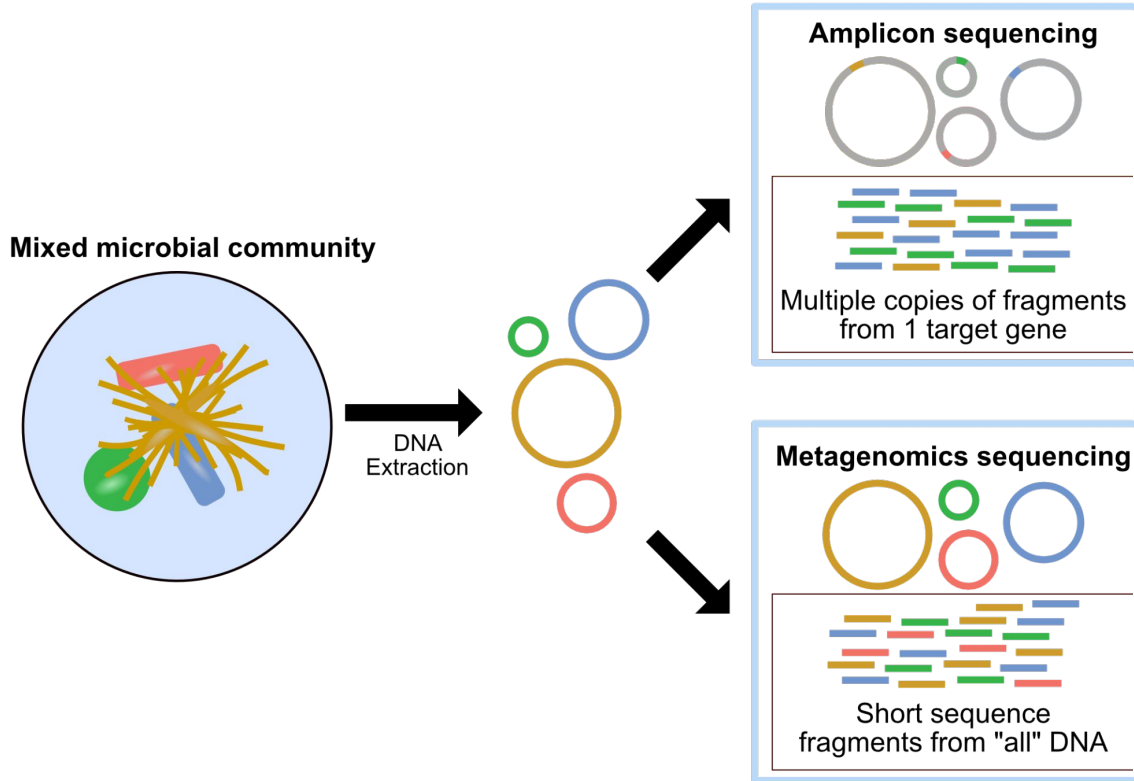
The goal is to completely characterise a microbial community:

- Who is there?
- What are they doing?



Amplicon vs metagenomic sequencing

The goal is to completely characterise a microbial community:



Who is there?

What are they doing?
(+ who is there)

Quality control

Cleaning raw reads reduces the risk of contamination in downstream analyses.

You might want to consider:

- GC content (metaG)
- Duplicates (metaG)
- Trim reads/regions by quality score (metaG+amplicon)
- Sequence length
- Contamination from (metaG+amplicon):
 - Human/host: map samples against reference
 - Reagents: use negative controls
 - Sequencing process: clipping known primers/adaptor sequences



It is important to keep high quality controls throughout the whole workflow

Sequencing technologies

Illumina

- Short-reads (50-250 bp)
- 4-20k million reads per run
- Error rate <0.1%

Oxford Nanopore/PacBio SMRT

- Long-reads (>1 kbp)
- 20-100k reads per run
- Error rate 10-15%



PacBio HiFi

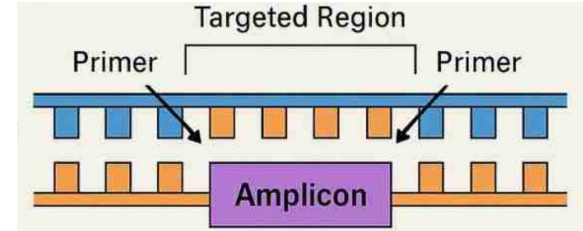
- Long-reads (up to 25-30 kb)
- Read accuracy ~99%

Amplicon

What is amplicon sequencing?

Any sequencing approach where a specific DNA region is PCR-amplified (an **amplicon**) and then sequenced

- Very cheap and quicker to run



Metabarcoding uses DNA barcodes, which are standardised genetic markers amplified with universal primers that work across taxa

- Great for population studies



Universal taxonomic markers

We want DNA markers that:

- Are conserved enough across various organisms
- Are variable enough to determine the organism of origin

What type of machinery could match this description? Is there a molecular structure or mechanism that is common across a multitude of organisms with distinct DNA markers?

Universal taxonomic markers

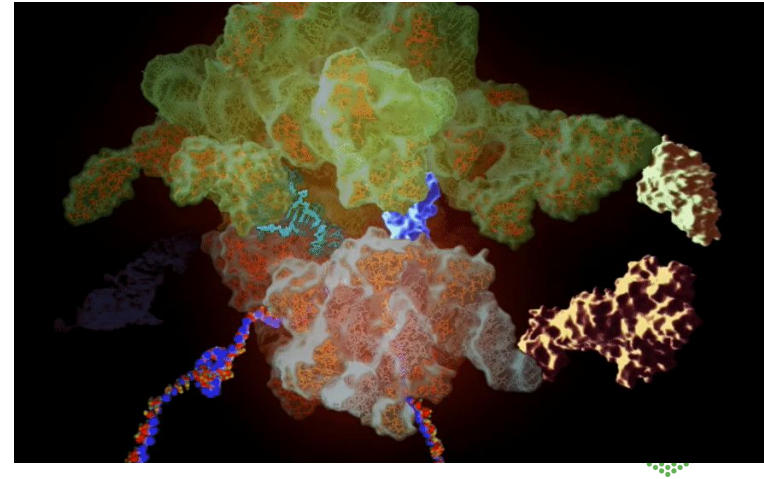
We want DNA markers that:

- Are conserved enough across various organisms
- Are variable enough to determine the organism of origin

What type of machinery could match this description? Is there a molecular structure or mechanism that is common across a multitude of organisms with distinct DNA markers?

Yes, **Ribosomes**! The molecular structure that bind amino acids to messenger RNA (mRNA) chains to make proteins.

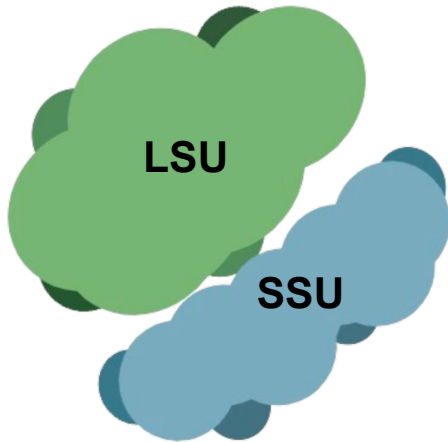
- It's a highly-conserved biological machinery, which makes them universally distributed
- It shows varying degrees of sequence variability, which allows them to be used as a species-specific signature



The 16S rRNA genes

- LSU: Large SubUnit
- SSU: Small SubUnit

Both made of proteins and RNA sequences

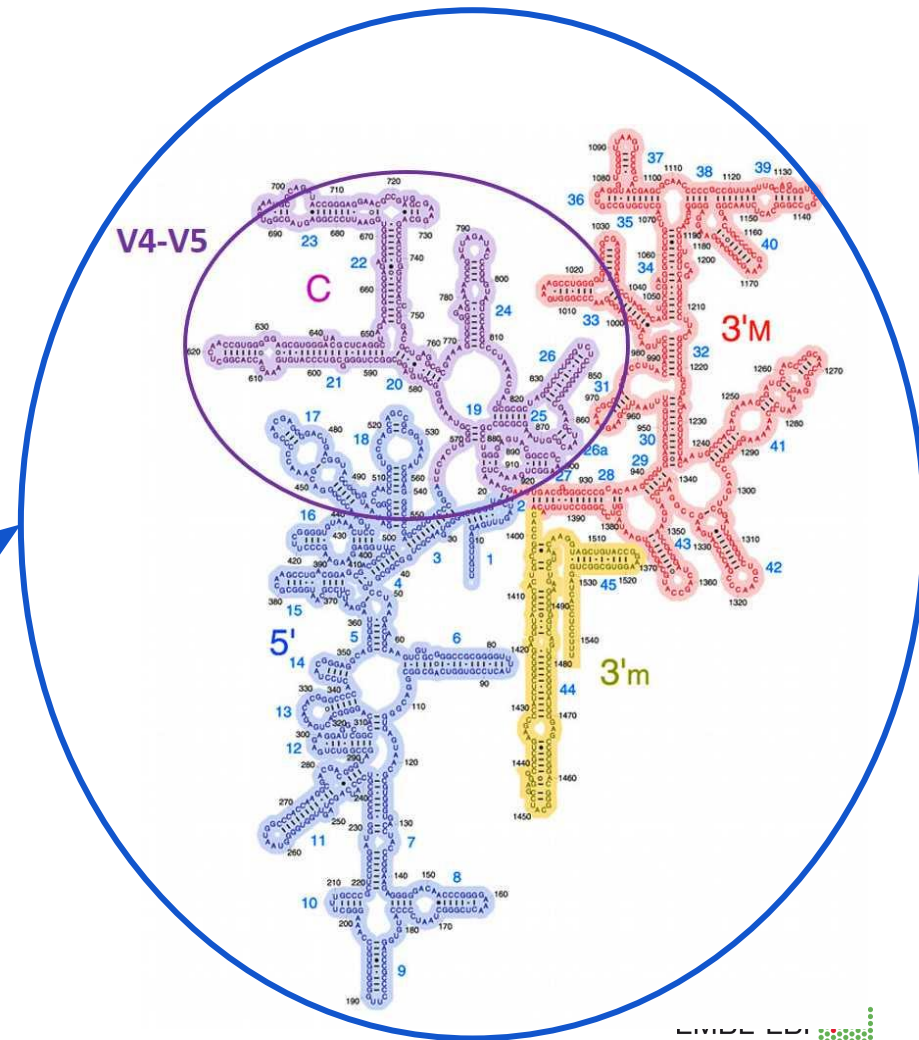
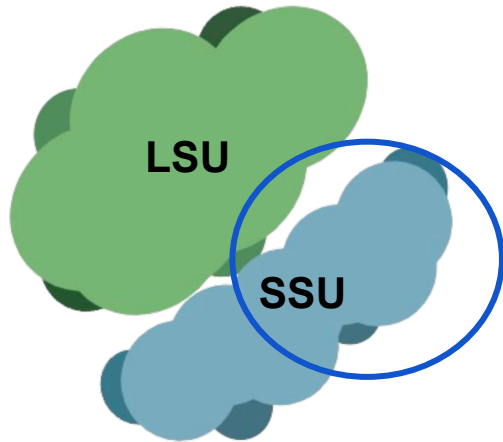


The 16S rRNA genes

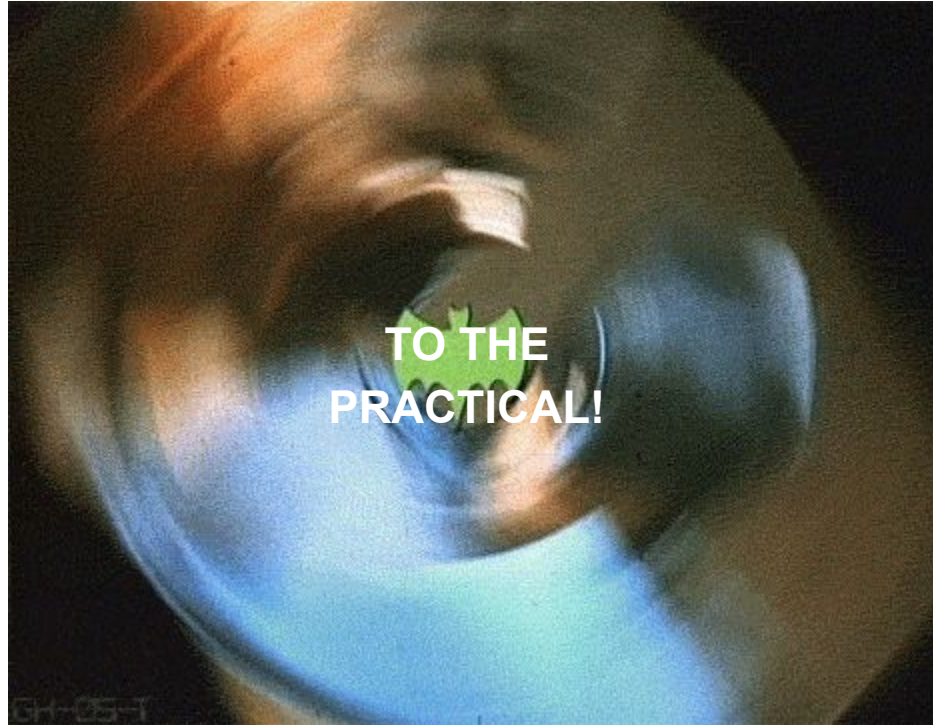
- LSU: Large SubUnit
- SSU: Small SubUnit

Both made of proteins and RNA sequences

The most widely used is the **16S rRNA**, as it is highly conserved between different species of bacteria and archaea



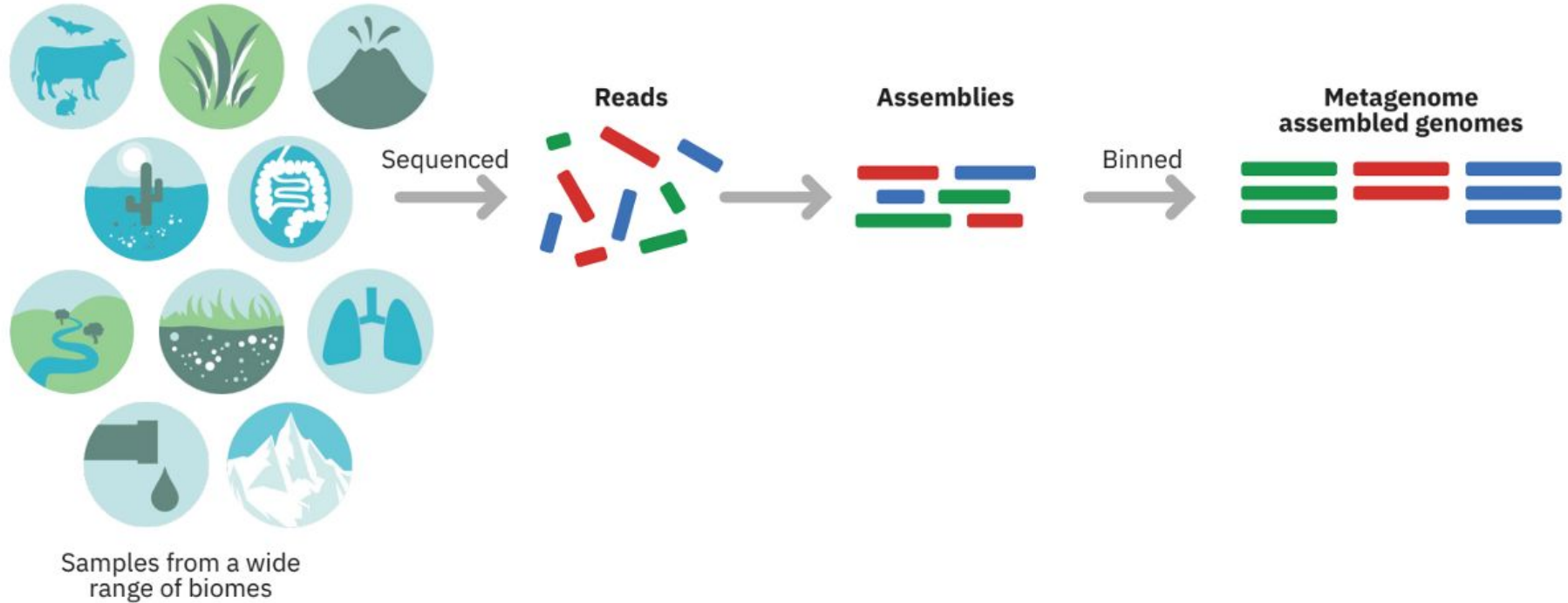
Amplicon Practical



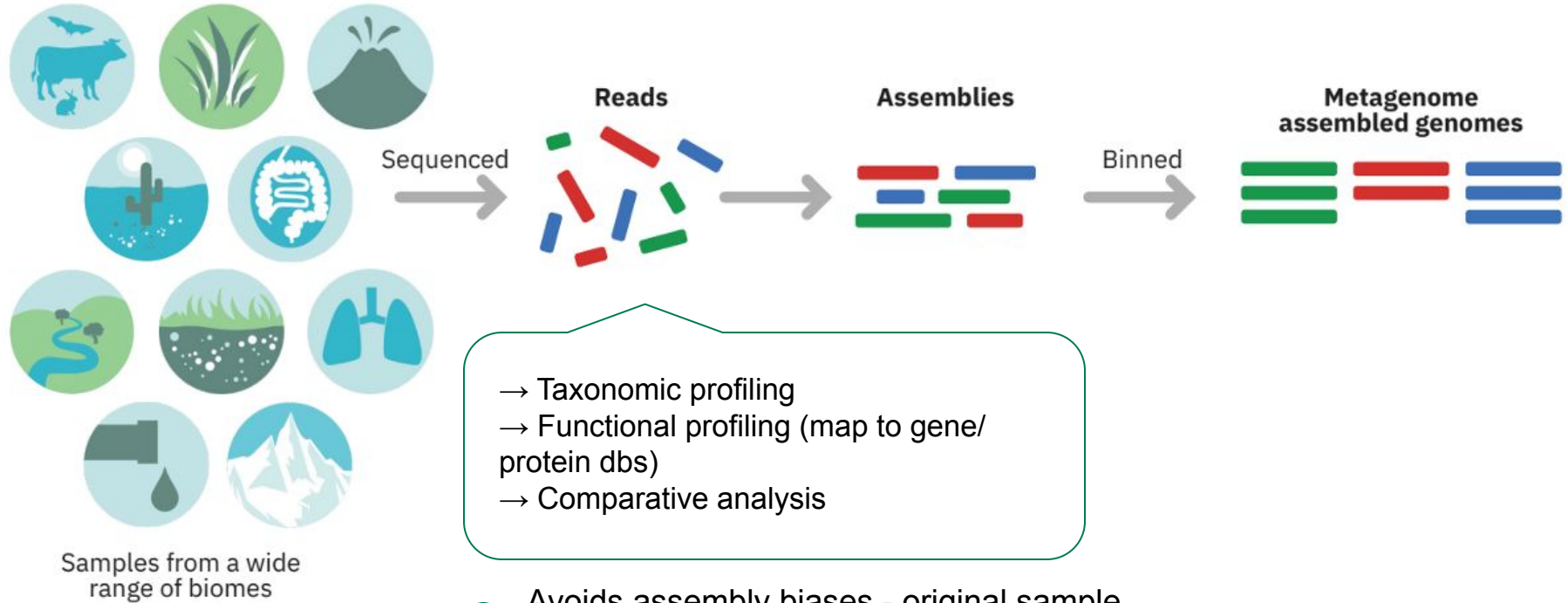
<http://plants-breeding.mgnify.org/sessions/amplicon.html>

Metagenomics

What can we do with metagenomic shotgun sequencing data?

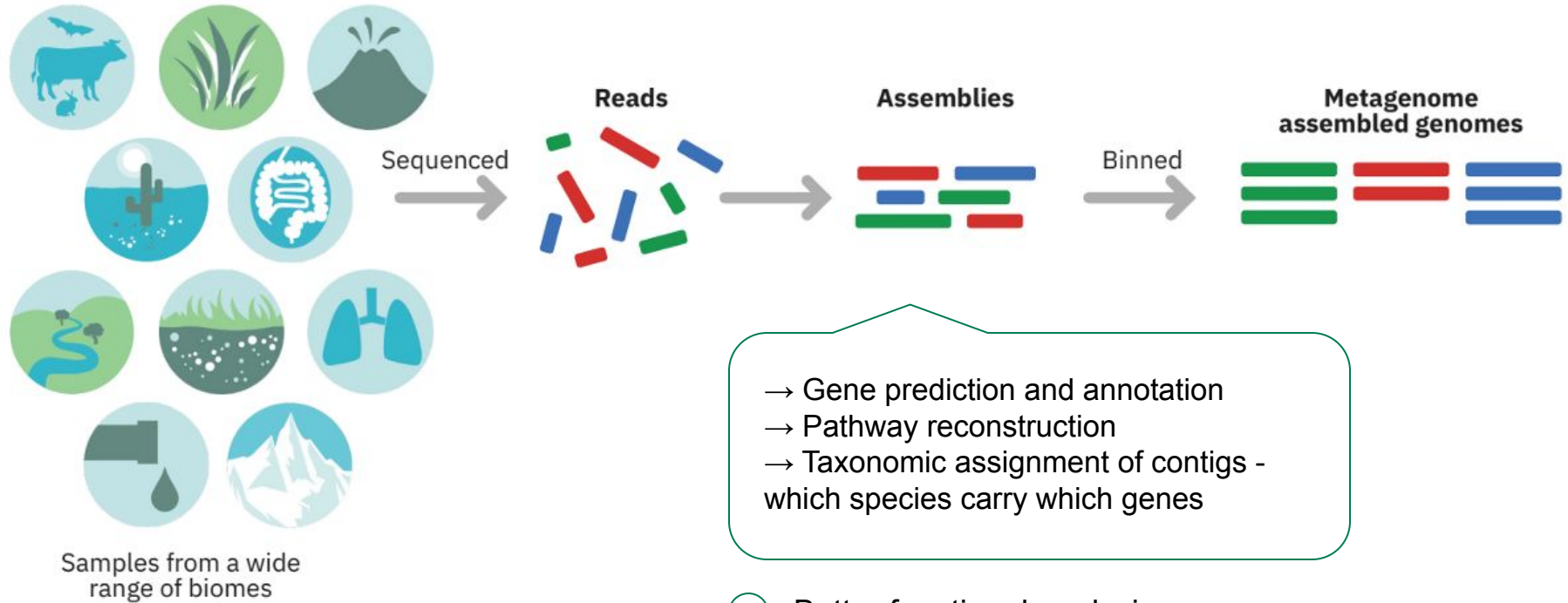


What can we do with metagenomic shotgun sequencing data?



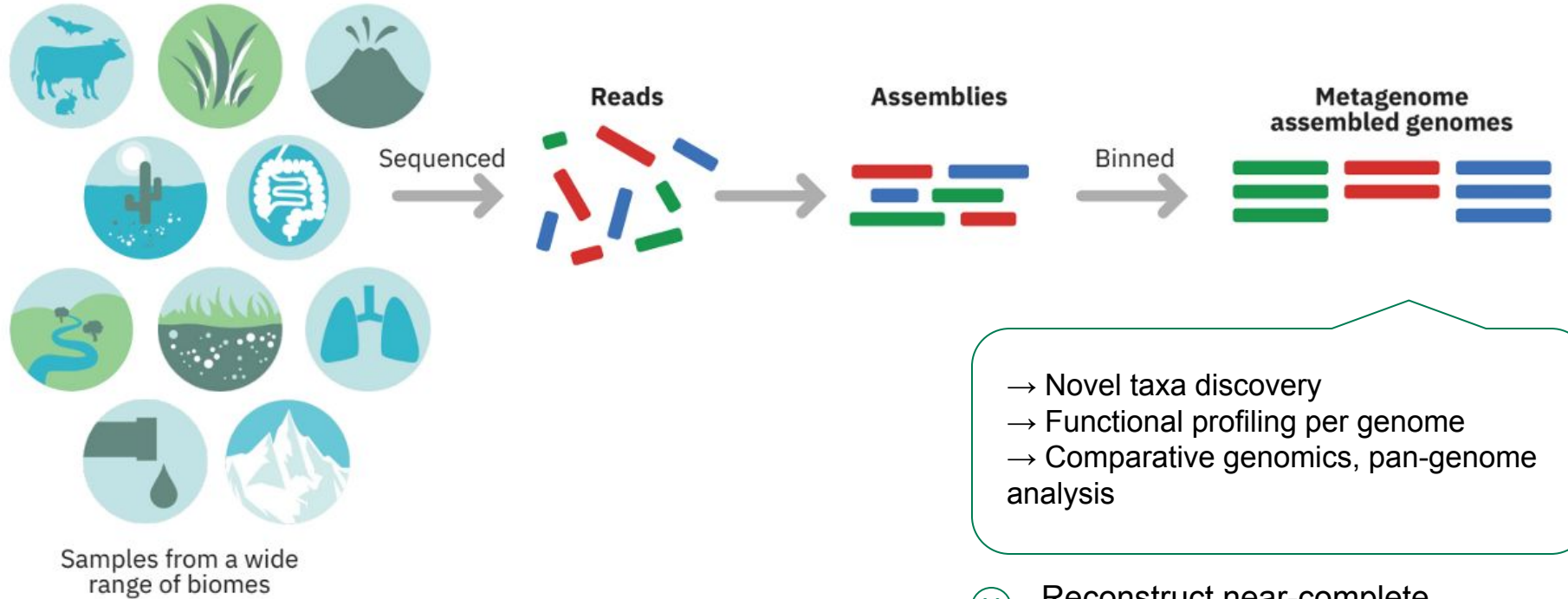
Avoids assembly biases - original sample composition is preserved

What can we do with metagenomic shotgun sequencing data?



Better functional analysis

What can we do with metagenomic shotgun sequencing data?



Reconstruct near-complete genomes

Why is metagenomics challenging?

- Vast number of sequences
- Most are missing from the reference databases
- Short sequence fragments are hard to characterise
- 'Metagenomics is like a disaster in a jigsaw shop'



- Millions of different pieces
- Thousands of different puzzles
- All mixed together
- Most of the pieces are missing
- No box art to refer to

Genomic vs metagenomic assembly

- Regular assembly assumes one genome with consistent sequence context.
- Don't know how many species are present and how closely related they are

Genomic vs metagenomic assembly

- Regular assembly assumes one genome with consistent sequence context.
- Approximately even genome coverage can be assumed
- Don't know how many species are present and how closely related they are
- Uneven coverage - some species are more abundant than others

Genomic vs metagenomic assembly

- Regular assembly assumes one genome with consistent sequence context.
- Approximately even genome coverage can be assumed
- Low variation
- Don't know how many species are present and how closely related they are
- Uneven coverage - some species are more abundant than others
- High strain-level variation

Genomic vs metagenomic assembly

- Regular assembly assumes one genome with consistent sequence context.
- Approximately even genome coverage can be assumed
- Low variation
- Don't know how many species are present and how closely related they are
- Uneven coverage - some species are more abundant than others
- High strain-level variation
- Different organisms can have shared and highly conserved sequences (e.g. rRNA genes) that don't assemble well

Genomic vs metagenomic assembly

- Regular assembly assumes one genome with consistent sequence context.
- Approximately even genome coverage can be assumed
- Low variation
- Don't know how many species are present and how closely related they are
- Uneven coverage - some species are more abundant than others
- High strain-level variation
- Different organisms can have shared and highly conserved sequences (e.g. rRNA genes) that don't assemble well
- Higher computational complexity (large assembly graphs)

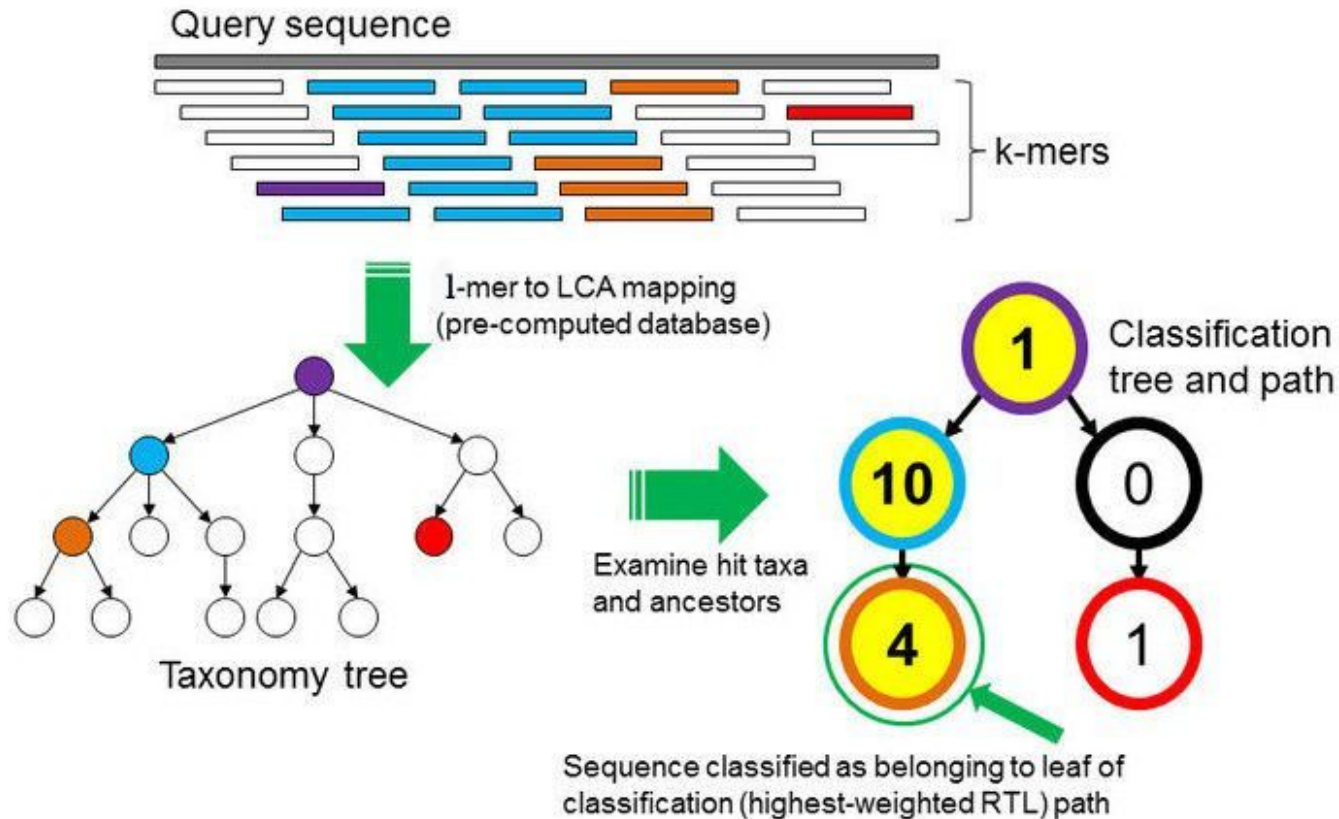
Genomic vs metagenomic assembly

- Regular assembly assumes one genome with consistent sequence context.
- Approximately even genome coverage can be assumed
- Low variation
- Don't know how many species are present and how closely related they are
- Uneven coverage - some species are more abundant than others
- High strain-level variation
- Different organisms can have shared and highly conserved sequences (e.g. rRNA genes) that don't assemble well
- Higher computational complexity (large assembly graphs)
- Cannot validate genomes without a reference

Concepts for the practical

- We will be using Kraken2
- Fast and optimised for efficient classification
- Very dependent on the database used
- No alignment, no model learning, only exact matches to the database

Kraken2 algorithm



Alpha Diversity: Within-Sample Diversity

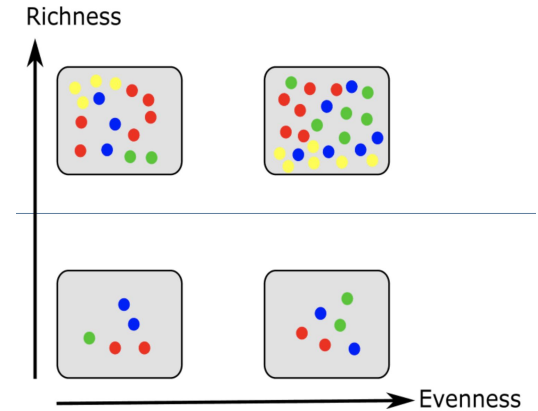
Measures diversity within a single sample

Captures:

- Richness (number of taxa)
- Evenness (distribution of abundances)

Sensitive to:

- Sequencing depth
- Database completeness
- Taxonomic resolution (species vs genus vs family)



Alpha Diversity: Within-Sample Diversity

We will assess alpha diversity by computing **Shannon diversity index**:

$$H = - \sum_i p_i \log(p_i)$$

Here, p_i is the proportion (relative abundance) of taxon i in the sample

Alpha Diversity: Within-Sample Diversity

We will assess alpha diversity by computing **Shannon diversity index**:

$$H = - \sum_i p_i \log(p_i)$$

Here, p_i is the proportion (relative abundance) of taxon i in the sample

Scenario 1: Two species, equal abundance

Species	Reads
A	50
B	50

$$H = -(0.5 \log 0.5 + 0.5 \log 0.5) = -2 \times (0.5 \times -0.6931) = 0.6931$$

Alpha Diversity: Within-Sample Diversity

We will assess alpha diversity by computing **Shannon diversity index**:

$$H = - \sum_i p_i \log(p_i)$$

Here, p_i is the proportion (relative abundance) of taxon i in the sample

Scenario 1: Two species, equal abundance

Species	Reads
A	50
B	50

Low richness (2 species)
but perfect evenness →
moderate H .

$$H = -(0.5 \log 0.5 + 0.5 \log 0.5) = -2 \times (0.5 \times -0.6931) = 0.6931$$

Alpha Diversity: Within-Sample Diversity

We will assess alpha diversity by computing **Shannon diversity index**:

$$H = - \sum_i p_i \log(p_i)$$

Here, p_i is the proportion (relative abundance) of taxon i in the sample

Scenario 2: Two species, uneven abundance

Species	Reads
A	90
B	10

$$H = -(0.9 \log 0.9 + 0.1 \log 0.1) \approx 0.325$$

Alpha Diversity: Within-Sample Diversity

We will assess alpha diversity by computing **Shannon diversity index**:

$$H = - \sum_i p_i \log(p_i)$$

Here, p_i is the proportion (relative abundance) of taxon i in the sample

Scenario 2: Two species, uneven abundance

Species	Reads
A	90
B	10

Low richness (2 species)
but uneven distribution
→ lower H .

$$H = -(0.9 \log 0.9 + 0.1 \log 0.1) \approx 0.325$$

Alpha Diversity: Within-Sample Diversity

We will assess alpha diversity by computing **Shannon diversity index**:

$$H = - \sum_i p_i \log(p_i)$$

Here, p_i is the proportion (relative abundance) of taxon i in the sample

Scenario 3: Many species, even abundance

$$H = -5 \times (0.2 \log 0.2) \approx 1.609$$

Species	Reads
A	10
B	10
C	10
D	10
E	10

Alpha Diversity: Within-Sample Diversity

We will assess alpha diversity by computing **Shannon diversity index**:

$$H = - \sum_i p_i \log(p_i)$$

Here, p_i is the proportion (relative abundance) of taxon i in the sample

Scenario 3: Many species, even abundance

$$H = -5 \times (0.2 \log 0.2) \approx 1.609$$

Higher richness $\rightarrow$ higher H

Species	Reads
A	10
B	10
C	10
D	10
E	10

Alpha Diversity: Within-Sample Diversity

We will assess alpha diversity by computing **Shannon diversity index**:

$$H = - \sum_i p_i \log(p_i)$$

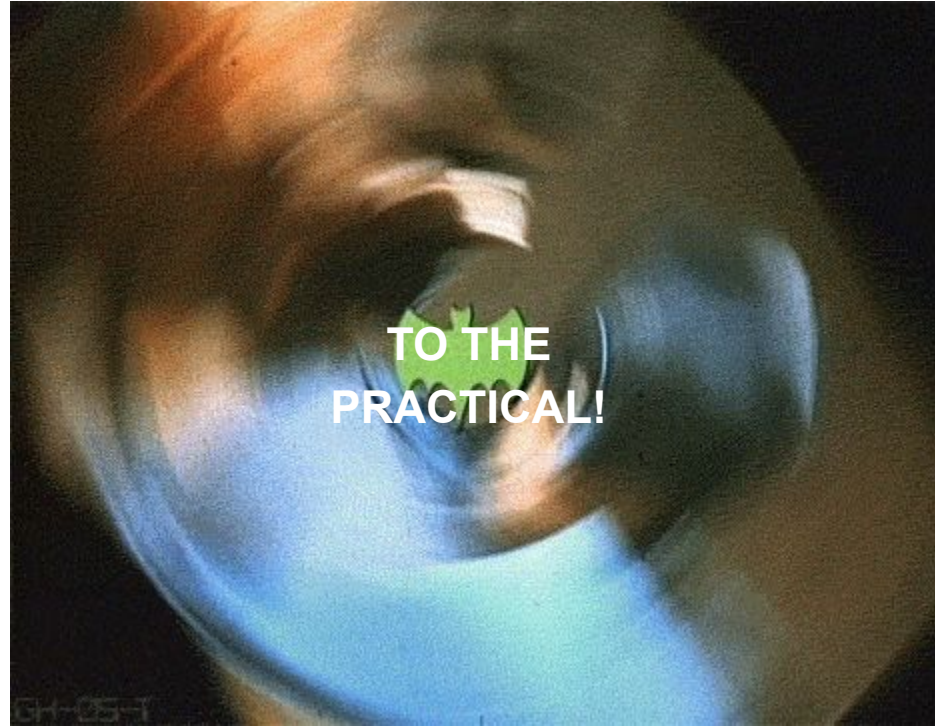
Here, p_i is the proportion (relative abundance) of taxon i in the sample

Shannon index increases with both richness and evenness.

Low Shannon: Few taxa or one dominant taxon

High Shannon: Many taxa, roughly equal abundance

Metagenomics Practical



<http://plants-breeding.mgnify.org/sessions/metagenomics.html>

The MGnify Resource

Why is there a need for a metagenomic resource?

Reproducibility and comparability of results

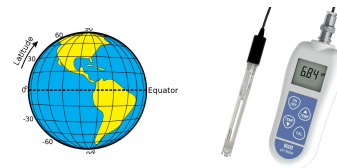
Reproducibility and comparability of results

- No perfect tool for analysis
- Different analysis tools can give different results
- The same tools can give different results depending on the version and underlying algorithm (e.g., HMMER2 vs HMMER3)
- The same version of the same tools can give different results depending on the reference database used

Longevity and usefulness of data

For data to have longevity and be useful to the scientific community, sequences need to be **archived** with **contextual metadata**

- How was it sampled?
 - How was it extracted?
 - How was it stored?
 - What sequencing platform was used?
-
- Where did it come from?
 - What were the environmental conditions (lat/long, depth, pH, salinity, temperature...)?
 - What were the clinical observations?



Contextual data

If contextual data is adequately described, querying and interpretation across projects becomes possible

- What microbial species are found in the North Pacific
 - ... at depths of 50 - 100 meters
 - ... in samples taken in May-June
 - ... compared to the Indian Ocean under the same conditions



Our analysis service

MGnify

- We offer a powerful metagenomic data analysis with a **versioned analysis pipeline** (currently v5.0)
- Our analysis is **biome agnostic**
- Data archiving with **contextual metadata**
- Allowing anyone in the world to **access** data and/or have their data **analysed** by the pipeline **free of charge**
- **Public** and **private** data analysis

Data resource

Analysis service

The screenshot shows the MGnify website interface. A green arrow points from the 'Data resource' label to the 'Text search' button in the 'Search by' section. Another green arrow points from the 'Analysis service' label to the 'Request' button in the 'Request analysis of' section.

MGnify
Submit, analyse, discover and compare microbiome data

Search MGnify

Example searches: Tara oceans, MGYS00000410, Human Gut

Overview Submit data Text search Sequence search Browse data About Help Login

Search by

- Text search**
Name, biome, or keyword
- Sequence search**
Sequence search

Or by data type

Analysis types		Public data	
495889	amplicon	5069	studies
60113	assemblies	615321	analyses
2050	metabarcoding	481714	genomes in 12 MAG catalogues
40094	metagenomes		
2581	metatranscriptomics		
2	long reads assemblies		

Or by selected biomes

Request analysis of

- Submit and/or Request**
Your data
- Request**
A public dataset

Latest studies

- EMG produced TPA metagenomics assembly of the Human Microbiome Project (HMP) Metagenomic WGS Project...**
The human metagenome Third Party Annotation (TPA) assembly was derived from the primary whole genome shotgun (WGS) data set, (PRJNA48479). This project includes samples from the following biomes: Fecal, Oral, Attached Keratinized gingiva, buccal mucosa...
- EMG produced TPA metagenomics assembly of PRJNA230567 data set (Systems Biology of Lipid Accumulation)**
The Third Party Annotation (TPA) assembly was derived from the primary whole genome shotgun (WGS) data set PRJNA230567, and was assembled with metaspades v3.15.3. This project includes samples from the following biomes: root:Engineered:Wastewater.

Data and analysis types in MGnify

	Taxonomic Analysis	Functional analysis	Extended functional analysis
Amplicon			
Raw reads (metagenomic or metatranscriptomic)			
Assembly			

(Raw reads are now rarely analyzed)

Demo on MGnify



Group discussion task

Use what you learned so far today to discuss the following:

Let's say you are planning a study of a plant microbiome.

How will you decide if you should do amplicon or metagenomic sequencing?

What factors are you going to consider?

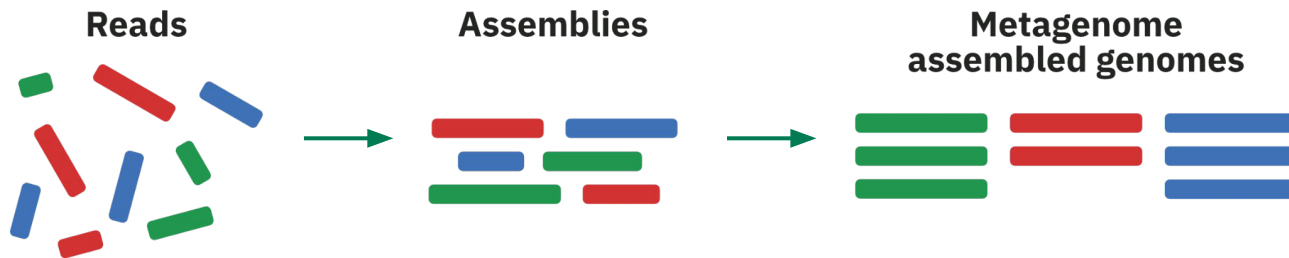
Put together a list to share with the group

Future Directions

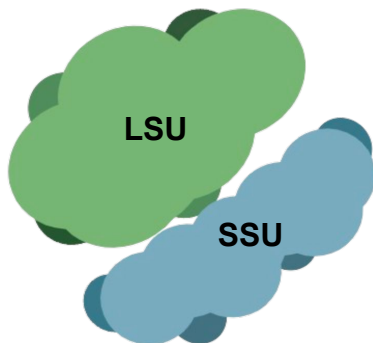
Advantages and shortcomings of metagenomics
and amplicon data

Microbiome studies: isolates are not enough!

- Metagenomics

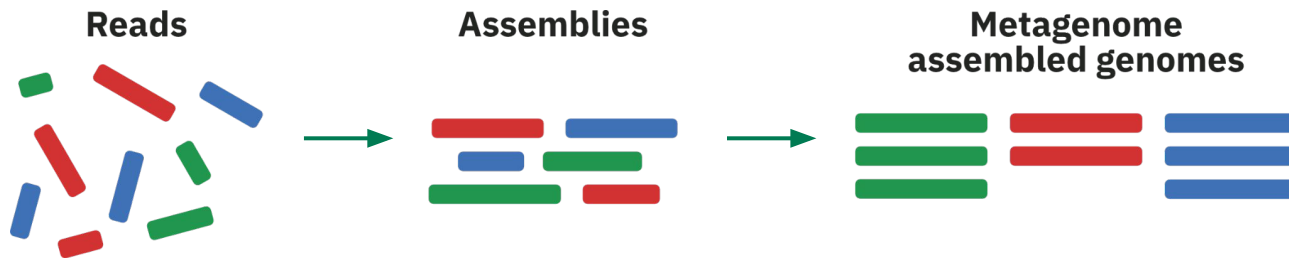


- Metabarcoding (amplicon)

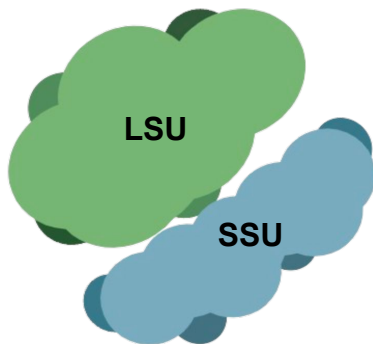


Microbiome studies: isolates are not enough!

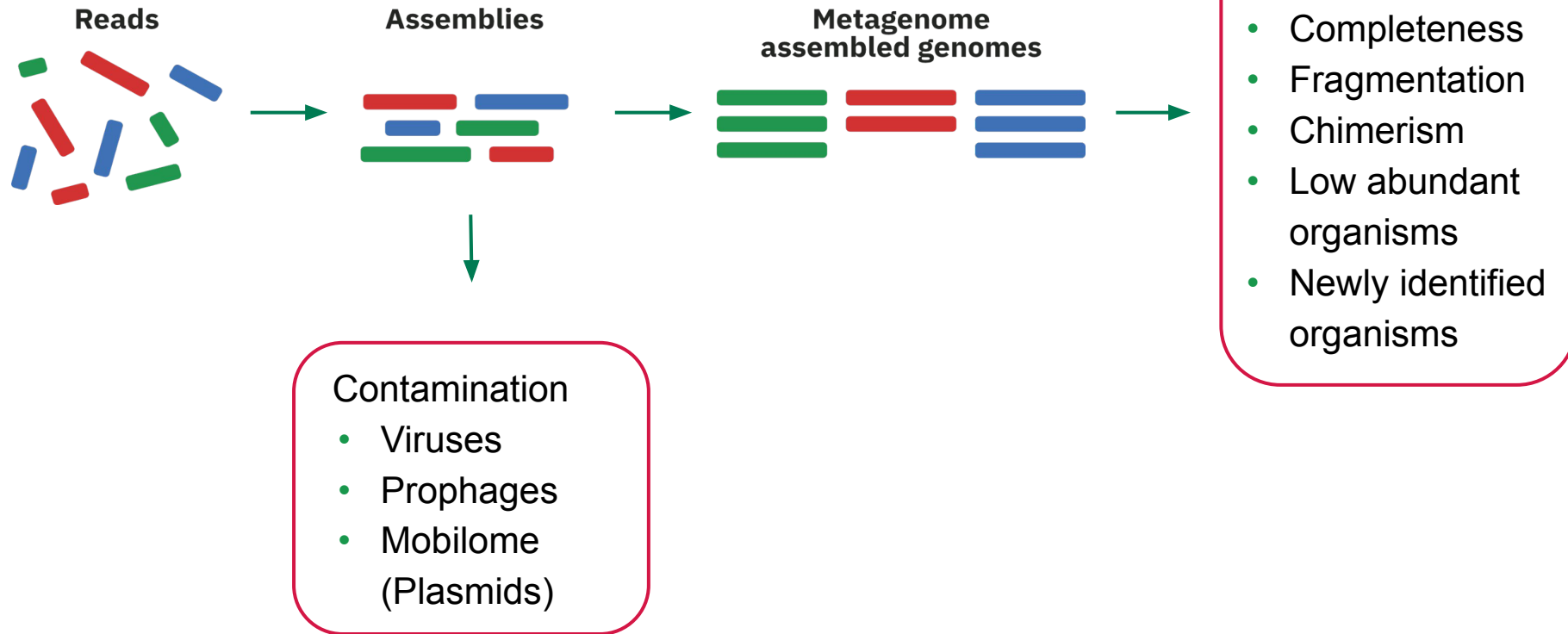
- Metagenomics



- Metabarcoding (amplicon)



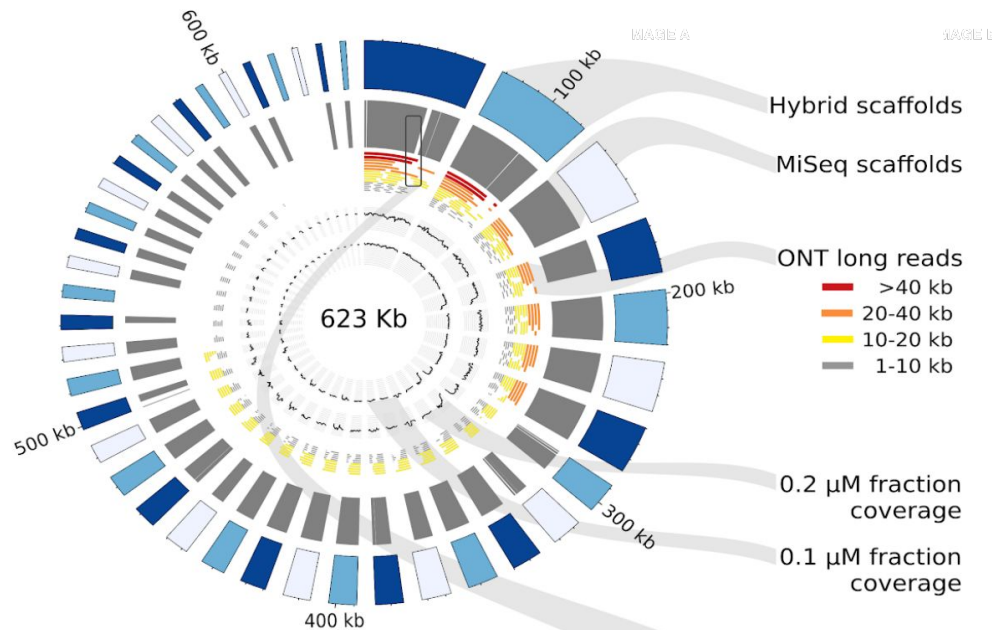
Metagenomics - short reads



Metagenomics - long reads

- Contamination
- Completeness
- Fragmentation
- Chimerism
- Low abundant organisms
- LARGE DNA fragments needed

Overholt, W. A. et al., Environmental Microbiology (2020)



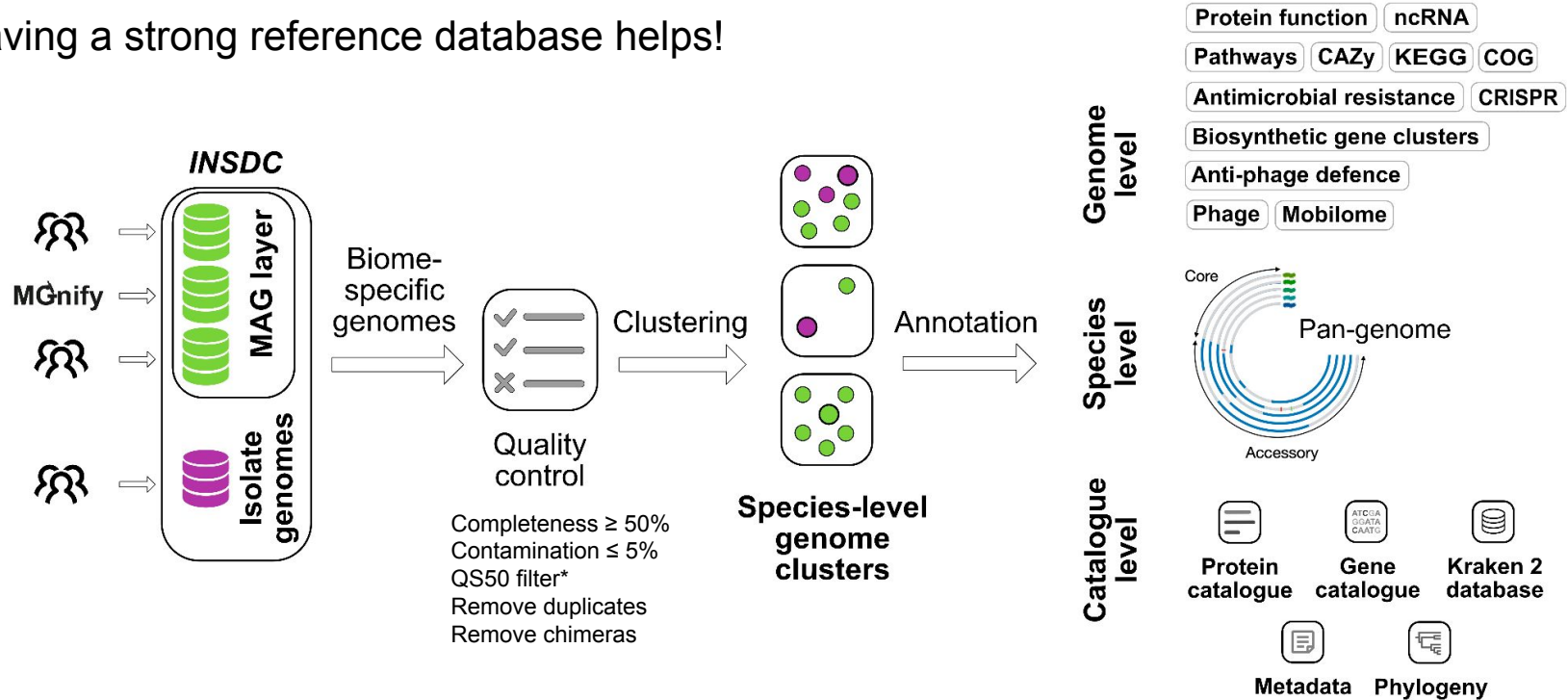
Mean coverage: 121X / 904X

Long reads spanning a
MiSeq scaffold gap |



Metagenomics - genome catalogues

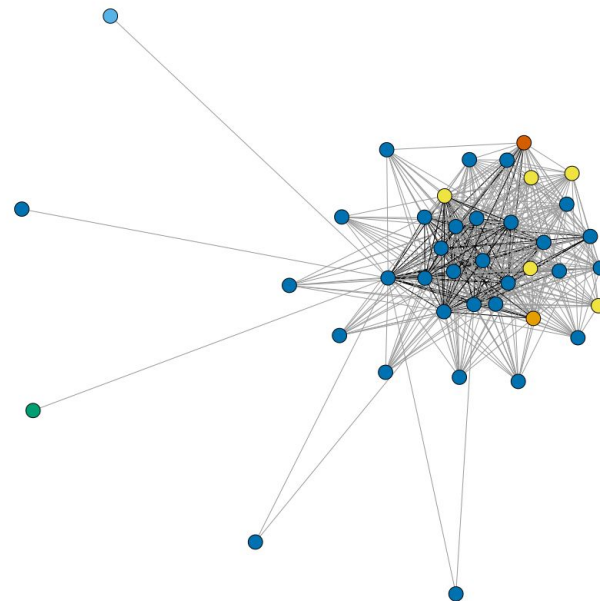
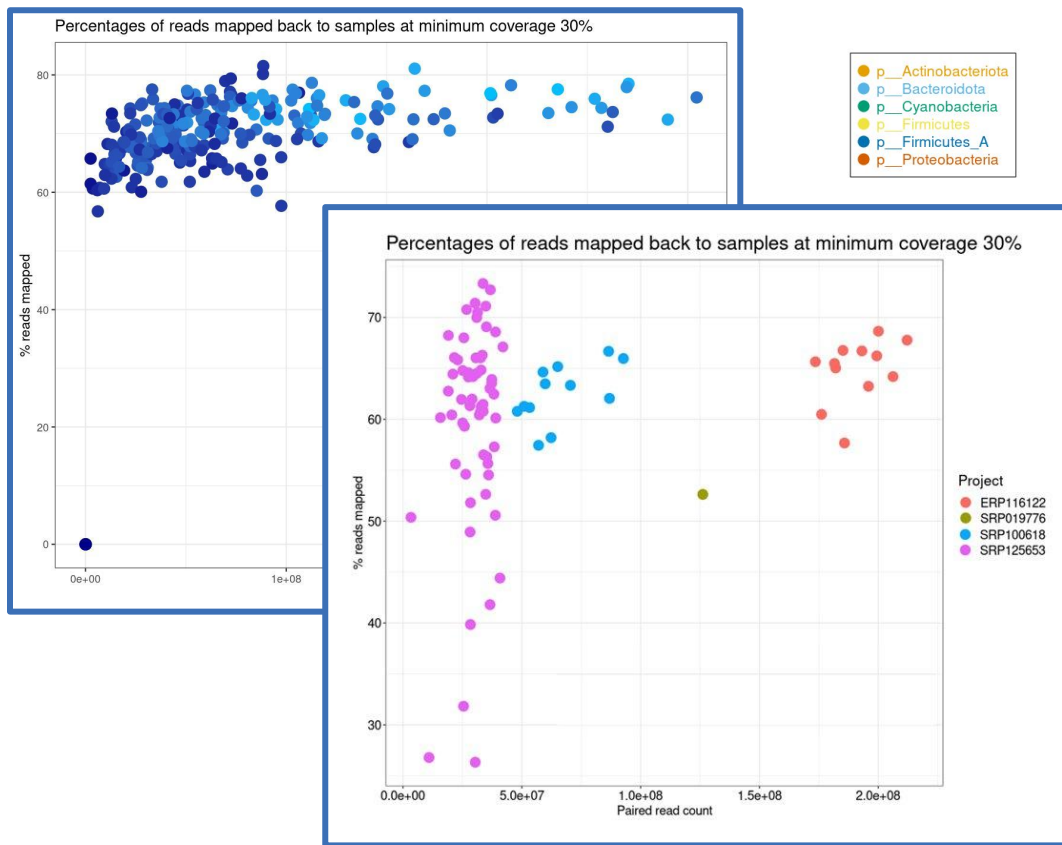
Having a strong reference database helps!



QS = %completeness - 5 x % contamination

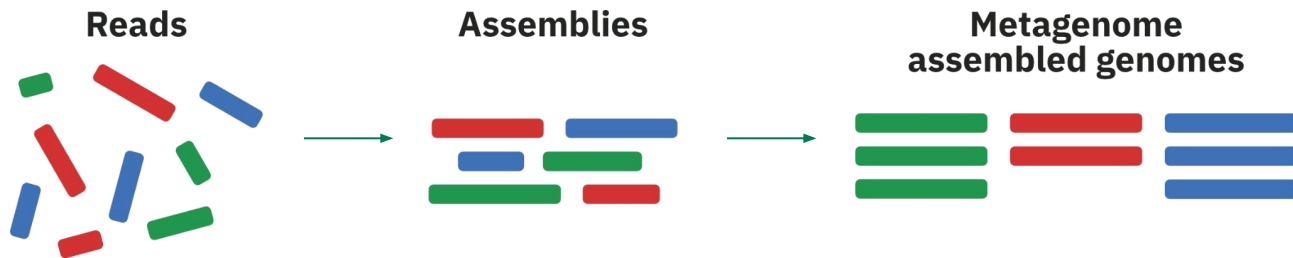
Download data from website, FTP, API

Mapping samples vs catalogues

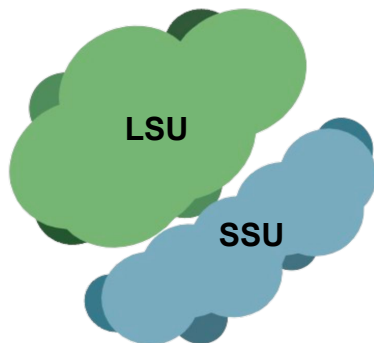


Microbiome studies: isolates are not enough!

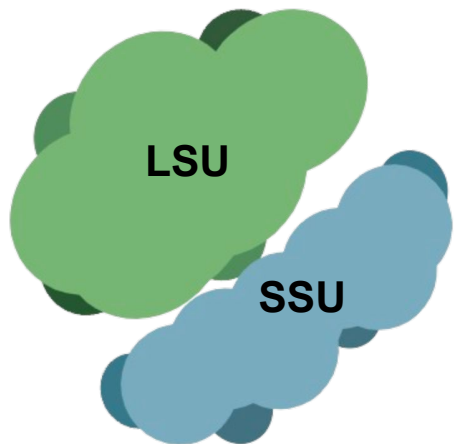
- Metagenomics



- Metabarcoding (amplicon)



Metabarcoding shortcomings

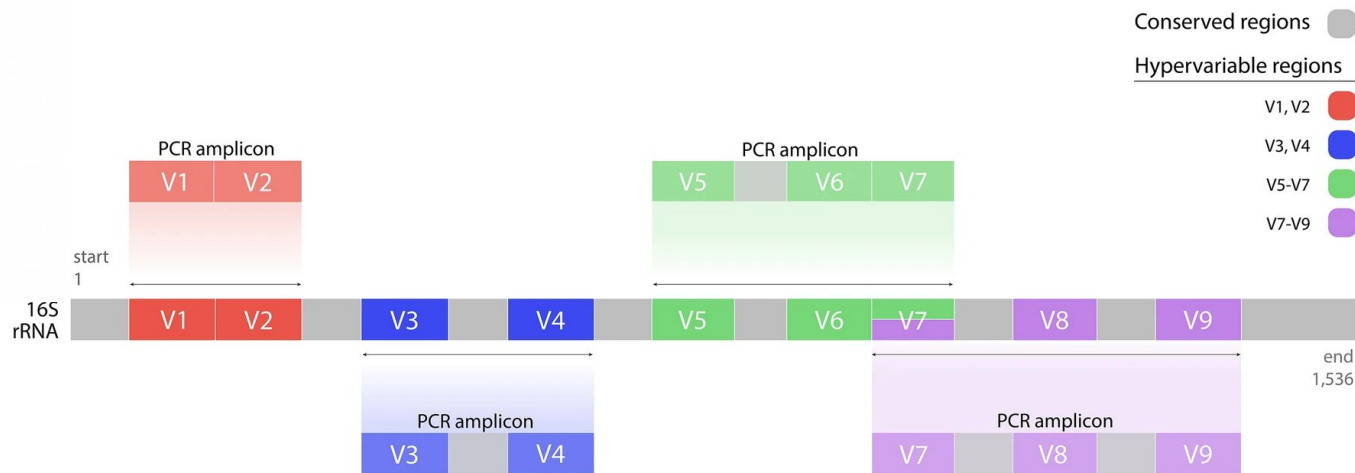
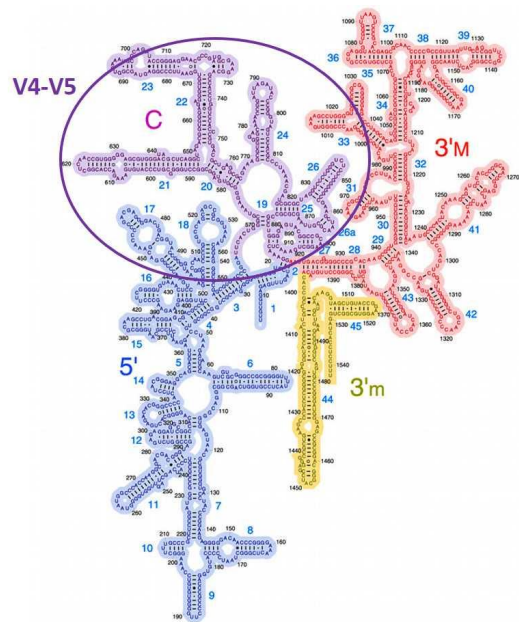


- Different regions selected
- Uncomparable across variable regions
- Taxonomy resolution
- Precision (OTUs)
- CNVs

Region selection

You don't usually sequence the whole 16S rRNA, but just a few regions.

An extra complication in metabarcoding sequencing: depending on the selected regions interval, results might differ (e.g. V1 vs V7-V9)



Metabarcoding - ASV detection

On top of LSU, SSU, ITS annotation:

- Different regions selected
- Uncomparable across variable regions
- Taxonomy resolution
- Precision
- CNVs



- **Amplified region inference** (if lacking metadata)
- **Primer** identification, prediction, trimming, and validation
 - **Sequence primers** information crucial for comparative studies and for **ASV calling**
 - Not removing primers can lead to **chimeric ASVs**
- **Read-ASV mapping:** necessary for **downstream analysis** of runs with more than one amplified region

Sample comparison and global context

Some issues are common to both metabarcoding and metagenomics

- Relative abundance
- Taxonomic resolution / novelty
- Low biomass/abundant organisms

Mapping vs strong references (with metagenomics and metabarcoding):

- Genome catalogues
- Culture collections
- Presence/absence of organisms across datasets

Can we see the same communities when comparing amplicon and metagenomic analyses of the same sample?

FindingPheno

Database of analyses data from maize, barley, soybean, tomato, and wheat in the FindingPheno project.

This data is available through a super study on the MGnify website.

[Home](#) > [Super Studies](#) > findingpheno

Super Study

FindingPheno

FindingPheno is creating an integrated computational framework for hologenomic big data, providing the tools to better understand how host-microbiome interactions can affect growth and other outcomes. This super study pulls together analyses of microbiome datasets that are relevant to the FindingPheno project.

▼ Flagship Projects

Biome	Study accession	Name	Abstract	Samples Count	Last Updated
	MGYS00006231	EMG produced TPA metagenomics assembly of PRJNA789467 data set (Disentangling the genetic basis of rhizosphere microbiome assembly in tomato (metagenomics part)).	The Third Party Annotation (TPA) assembly was derived from the primary whole genome shotgun (WGS) data set PRJNA789467, and was assembled with metaSPAdes v3.15.3. This project includes samples from the following biomes: root:Host-associated:Plants:Rh...	114	05/07/2025
	MGYS00001034	Changes in the bacterial communities of subtropical soils from Argentina with different land uses.	The goal of this study was to analyze the modifications in the structure of the bacterial communities from Argentinean subtropical soils caused by changes in land use. The samples included soils from undisturbed forests and from sites that were defor...	19	12/03/2024



MICROBE

A collection of studies on soil and seeds datasets to identify *keystone taxa* and understand how to better cryopreserve and isolate them.

[Home](#) > [Super Studies](#) > MICROBE

Super Study

MICROBE

MICROBE paves the way for an innovative microbiome biobanking in Europe by combining scientific expertise with research infrastructures know-how.

More info at <https://www.microbeproject.eu/>

► Flagship Projects

► Related Projects

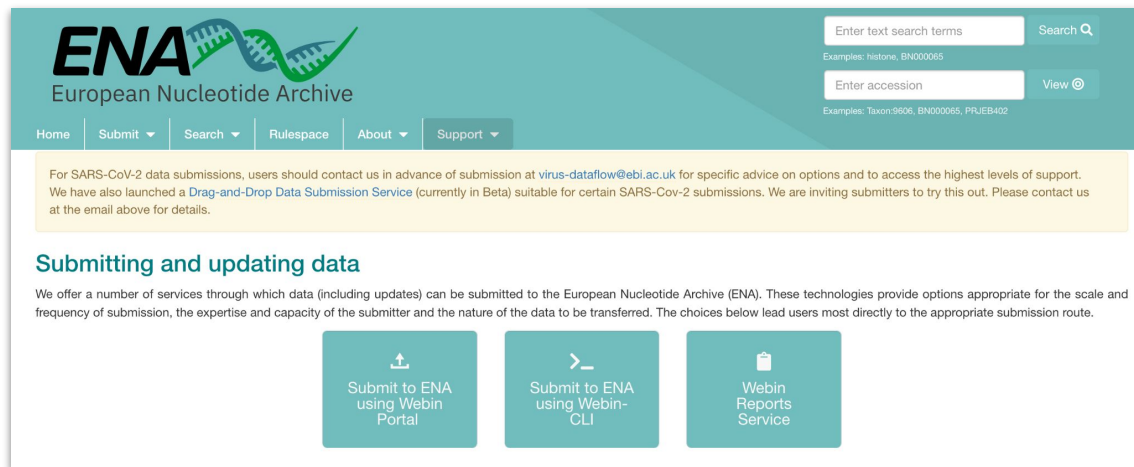
Biome	Study accession	Name	Abstract	Samples Count	Last Updated
	MGYS00006821	16S rRNA analysis of bacterial communities of forest and grassland sites located in the German Biodiversity Exploratory Schwäbische Alb	Soil bacteria are important components for nearly all biogeochemical cycles in terrestrial ecosystems and participate in most nutrient transformations in soil. In contrast to the importance of soil bacteria for ecosystem functions, we understand little how different management types affect the soil bacterial community composition. We used pyrosequencing-based analysis of the V2-V3 16S rRNA gene region to identify changes in bacterial diversity and community structure in nine forest and nine grassland soils under six different management types (three replicates each) from the Schwäbische Alb (southwestern Germany). The data set comprised 598,962 sequences that were affiliated to the domain Bacteria. The number of classified sequences per sample ranged from 23,515 to 39,259. The dominant taxonomic groups across all samples (> 1% of all sequences) were Acidobacteria, Alphaproteobacteria, Actinobacteria,	18	02/07/2025



A note on data submission

Resources such as MGnify are made possible when users submit their data and associated metadata to the public archives

We encourage you to submit your data to INSDC and provide as much metadata as possible.



The screenshot shows the ENA website header with the logo and navigation menu. The main content area features a yellow banner with information about SARS-CoV-2 submissions, a section titled 'Submitting and updating data' with explanatory text, and three teal buttons for submission methods: 'Submit to ENA using Webin Portal', 'Submit to ENA using Webin-CLI', and 'Webin Reports Service'.

ENA
European Nucleotide Archive

Home | Submit | Search | Rulespace | About | Support

Enter text search terms Search

Examples: Histone, BN000065

Enter accession View

Examples: Taxon:9606, BN000065, PRJEB402

For SARS-CoV-2 data submissions, users should contact us in advance of submission at virus-dataflow@ebi.ac.uk for specific advice on options and to access the highest levels of support. We have also launched a [Drag-and-Drop Data Submission Service](#) (currently in Beta) suitable for certain SARS-Cov-2 submissions. We are inviting submitters to try this out. Please contact us at the email above for details.

Submitting and updating data

We offer a number of services through which data (including updates) can be submitted to the European Nucleotide Archive (ENA). These technologies provide options appropriate for the scale and frequency of submission, the expertise and capacity of the submitter and the nature of the data to be transferred. The choices below lead users most directly to the appropriate submission route.

- Submit to ENA using Webin Portal
- Submit to ENA using Webin-CLI
- Webin Reports Service

Acknowledgements

Rob Finn (MGnify Group Leader)

Lorna Richardson

Chris Atallah

Martin Beracochea

Manuel Carbajo

Alejandra Escobar

Santiago Fragoso

Vikas Gupta

Vangelis Karatzas

Jennifer Mattock

Sofia Ochkalova

Aca Rajkovic

Sandy Rogers

Stephane Pesant

Tim Rozday

Kate Sakharova

Mahfouz Shehu

Disha Tandon

Johanna von Wachsmann



Tony Burdett
Dipayan Gupta

